

Establish basic Discharge Model

$$\frac{dSOC(t)}{dt} = -\frac{I(t)}{Q_{\text{eff}}}$$

Analysis of Influential factors of Power

Power-influencing Factor



Environmental Factors



Battery Characteristics



(a) Model Construction

OS Scheduling
Strategies

Advisories
for users

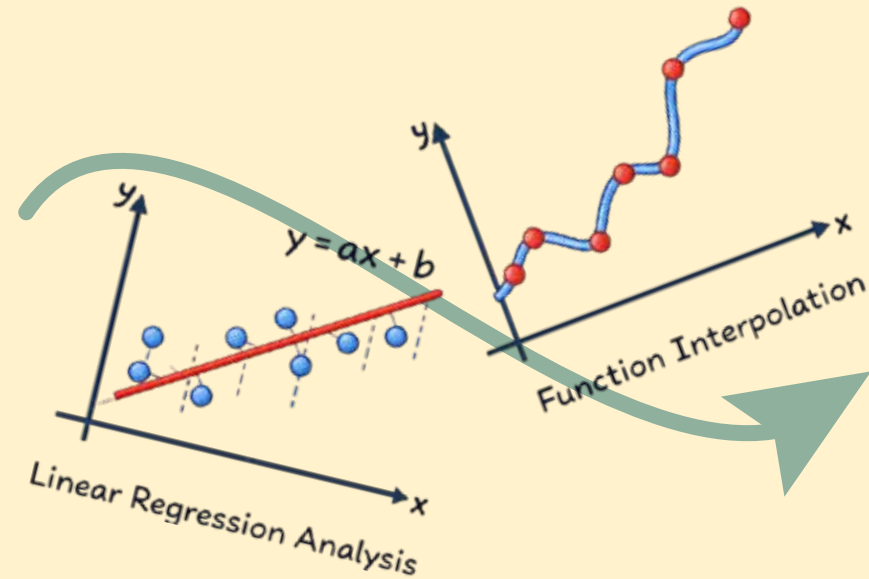
(d) Application

Opensource data

OCV-SOC
discharge curve

Cycletime-
capacity Curve

Debug data
from Cell Phone



(b) Data Processing

Model Parameters

Temperature-Cpacity
Coefficient

Capacity-reduction
Coefficient k

Power Params

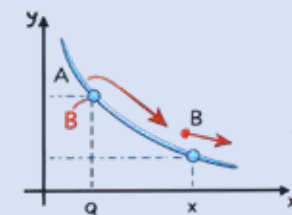
Time-to-Empty Estimations

Battery discharge
Simulations

Differential
Equation Solving

Sensitivity Analysis

Elasticity Analysis

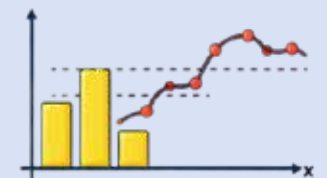


$$S_p = \frac{\partial \ln(\text{TTE})}{\partial \ln(p)}$$

Influential factor of battery life

Single-factor Perturbation Analysis

Graph
Visualizations



(c) Model Validation